

Sentinel SOC Detection & Response Lab

A hands-on Security Operations Center simulation — built to demonstrate practical threat detection and incident response skills

The Problem

Most cybersecurity job postings for SOC Analyst roles ask for hands-on experience with tools like Microsoft Sentinel, incident investigation, and threat detection — but breaking into the field without prior SOC experience makes that hard to demonstrate on a resume alone.

What I Built

I designed and built a complete, self-contained Security Operations Center lab in Microsoft Azure — simulating the full lifecycle of a real security team's work: detecting an attack, investigating it, and documenting the response.

Approach

- 1 **Set up the environment** — deployed a Windows Server virtual machine in Azure and connected it to Microsoft Sentinel (Microsoft's cloud-based security monitoring platform), so all system activity was logged and monitored in real time.
- 2 **Built custom detections** — wrote detection rules mapped to MITRE ATT&CK (the industry-standard framework for classifying attacker behavior), covering three real attack techniques: password brute-forcing, malicious PowerShell execution, and attacker persistence via scheduled tasks.
- 3 **Simulated real attacks** — safely launched each of these attacks against my own lab environment to test whether my detections actually worked.
- 4 **Investigated the results** — when Sentinel correctly flagged each attack as a security incident, I investigated it the way a SOC analyst would: reviewing the evidence, identifying what happened, and documenting the findings.
- 5 **Went hunting** — beyond automated alerts, I also proactively searched the logs for signs of attacker behavior not already covered by a detection rule (a practice called “threat hunting”).

Tools & Technologies

Microsoft Azure • Microsoft Sentinel • KQL (Kusto Query Language) • MITRE ATT&CK Framework • Windows Security Event Logs • Atomic Red Team (attack simulation)

Outcome

All three custom detections successfully identified their target attacks, generating real, investigable security incidents. Each incident was fully documented with evidence, root-cause analysis, and lessons learned — mirroring the deliverables expected of a working SOC analyst. The full technical project, including all detection code and incident reports, is publicly available on GitHub.

Why It Matters

This project shows I can go beyond studying security concepts in theory — I can build a monitored environment, write working detection logic, respond to real (simulated) attacks, and communicate findings clearly, which are the core day-to-day skills of a SOC Analyst.

GitHub: github.com/PathanZ/sentinel-soc-lab